

JYOTI SINHA

Mobile: 716-200-8183 | jyotisinhas21@gmail.com | [linkedin.com/in/jyoti-sinha20](https://www.linkedin.com/in/jyoti-sinha20) | github.com/jyotisinhaa

Software Engineer with 4+ years of experience building scalable microservices, agentic RAG systems, and high-performance backend APIs. Track record of measurable business impact through system optimization, cost reduction, and AI/ML-driven solutions in Java, Python, and Golang.

EXPERIENCE

SS&C Technologies, Software Engineer

California, United States
May 2022 - Current

- Redesigned the RAG pipeline from a basic v1 to an agentic RAG v2 (Python, LangChain, FastAPI), with Docling-based ingestion, hybrid chunking, and agentic retrieval for dynamic multi-step query resolution, driving full customer adoption of v2 and a 20% increase in user engagement.
- Extended a custom MCP architecture to Trino, building new tools for schema discovery and query access, with guardrails ensuring safe, production-grade responses.
- Replaced Starburst with open-source Trino, configured multi-database connectors and OPA-based access control, and optimized query execution and caching, achieving a 35% performance improvement and \$2M+ in annual cost savings.
- Architected Trino audit logging (QueryLog plugin, SS&C Cloud storage, Grafana integration), eliminating Kubernetes disk pressure and enabling centralized visibility.
- Developed a Java Spring Boot microservice with dynamic configuration, serving 50+ clients with self-service data retrieval from Trino via user-defined query parameters.
- Boosted system efficiency via pagination, streaming pipelines, and Redis caching, cutting response time by 21% and query execution by 30%, plus one-click CSV/Excel export for large-scale data.
- Built scalable microservices and RESTful/gRPC APIs across Java Spring Boot, C# ASP.NET, and Golang (SQLC, ProtoBuf), integrating with PostgreSQL and Starburst for enterprise backend systems.

BDIPLUS, Software Engineer

New York City, United States
Mar 2022 - May 2022

- Built a multi-step sign-up form (JavaScript, React, Redux, Bootstrap) with Node.js/SQL backend endpoints, shipping features and resolving critical bugs under tight deadlines.

INTERNSHIPS & RESEARCH EXPERIENCE

BDIPLUS, Data Engineer Intern

New York City, United States
May 2021 - Dec 2021

- Built a data pipeline extracting 4TB of data into HDFS using Apache Hive, PySpark, and Shell scripting; improved data accuracy 16% by revamping the record-matching algorithm.

University at Buffalo, Graduate Research Intern (Pioneer Research, Thesis)

Buffalo, United States
Aug 2020 - Dec 2021

- Built a cross-browser web app to collect and analyze user perception of biometric systems, using MongoDB, SPSS, and ML techniques (CNN, feature selection) to surface adoption concerns.

CORE COMPETENCIES

Programming Languages: Java, C#, Python, Golang, JavaScript, SQL, HTML, CSS, React JS, Bootstrap

AI & LLM Integration: LangChain, OpenAI, RAG, Docling, Hybrid Chunking, Vector DB (FAISS, Pinecone), MCP, Guardrails, Ollama, Hugging Face, Transformers, Prompt Engineering, TensorFlow, Keras, NumPy, Pandas, CNN, RNN

Continuous Learning: Model Fine-tuning (LoRA/QLoRA on Gemma, Llama), Quantization

Databases & Data Systems: PostgreSQL, MS SQL Server, MongoDB, Redis, Trino, Starburst, Apache Hive

Cloud, DevOps & Monitoring: Docker, Kubernetes, Helm, AWS (Lambda, EC2, S3), Model Deployment Pipelines, Grafana, Prometheus

Backend Frameworks & Tools: Spring Boot, Maven, Gradle, Hibernate, JPA, ASP.NET, FastAPI, RESTful APIs, gRPC, ProtoBuf, Streamlit

Machine Learning & Tools: TensorFlow, Keras, NumPy, Pandas, Clustering Algorithms, CNN, RNN, MLP, SPSS, Git, JIRA, Swagger, Postman

Core Strengths: System Design, Scalable Microservices, API Optimization, Data-Driven Problem Solving, Critical Thinking

EDUCATION

MS, Computer Science (AI Minor), University at Buffalo, USA

Aug 2019 - Feb 2022

Research Fellowship | Teaching Assistant (ML, Pattern Recognition, DBMS) | Perfect GPA

BE, Electronics and Telecommunication Engineering, BIT Mesra, India

Aug 2014 - May 2018

Best Outgoing Award